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Mathematical modelling

Maths is used to make sense of the world around us, and help us to make decisions, in our personal lives and in the worlds of business, politics, science, and so on. This could involve, for example:

- Creating and solving equations.
- Plotting graphs.
- Making calculations.
- Using computer software.

Each of these things could be thought of as creating a **mathematical model**. This chapter will introduce and explore a range of different kinds of model that may be useful, and when they may be encountered in the real world.

Quote

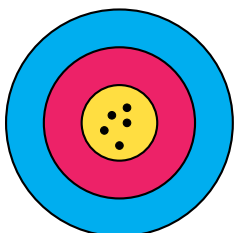
"All models are wrong, but some are useful."

George Box, Statistician

The above quote highlights that mathematical models don't always need to be able to produce 'the *right* answer', and in many situations such a thing is not even possible. Instead, the goal is often a simply a *useful* answer. Since we are entering a world of often *not knowing the true answer*, it is helpful to be learn some related vocabulary:

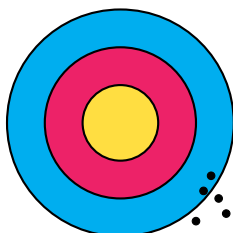
- **Accuracy** relates to how much a measurement or process leads towards the *true value*.
- **Precision** relates to how close repeated measurements or processes are to *each other*.

The results of four different archers aiming for the centre of a target should help illustrate the difference:



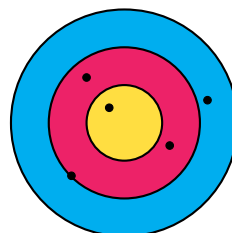
Accurate ✓

Precise ✓



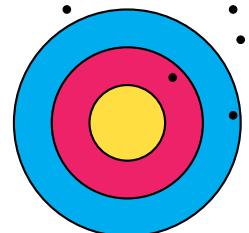
Accurate ✗

Precise ✓



Accurate ✓

Precise ✗



Accurate ✗

Precise ✗